Content Opportunity Finder

NTU Module 3 · Congrui Liu

Turns YouTube comments into 3 video ideas a week for AI creators

TARGET USER
AI YouTube creator
AI Career · AI Tools · AI Agents

PAIN POINT
Too many comments
today: ~3 h a week reading by hand

GOAL
3 video ideas a week
backed by real viewer questions

My idea: a video is worth making when the comments show 3 signals

DEMAND
Many people ask it
same question, many people

+

GAP
Few videos on it
asked, but no/few video yet

+

FRESHNESS
Rising this week
topic last week ↑ vs the weeks before

=

OPPORTUNITY
Worth filming now
score = 45% demand
+ 35% gap + 20% fresh

How it works: 5 steps from raw comments to 3 videos

51,447 comments
YouTube API · 120 days

→

25,226 clean
English, no spam

→

3,678 Y/N questions
Supervised: classify comments
L03 · L04 · L09

→

50 topics
Unsupervised: K-Means
L05 · L09

→

3 videos / week
Score + back-test
L06 · L09

Data: real, recent comments; half was noise; split by time

DATA 51,447 raw › 25,226 clean ▸ **FILTER** 493 labels › 356 train · 137 test › 3,678 questions ▸ **TOPICS** 2,574 Jul–Sep · 1,104 Oct › 50 topics ▸ **SCORE** 3 videos

1 Collect

1. **Official YouTube Data API**
   120 days (fresh enough) · 28 Jun – 26 Oct 2026
2. **3 niches = my 3 content lines**
   AI Career · AI Tools · AI Agents
3. **5 real search words each**
   e.g. "how to get a job in AI"

51,447 comments
1,282 videos · 893 channels
names hashed (HMAC*)

2 Clean: 51,447 → 25,226

Collected	51,447
English videos only langdetect (language library)*	40,805
At least 5 words	27,131
English comments langdetect*	25,658
Hindi-in-English removed word list*	25,424
Spam removed same text 3+ times	25,226

≈ half was noise · * beyond the course

3 Split by time (L06)

Cut on 28 Sep



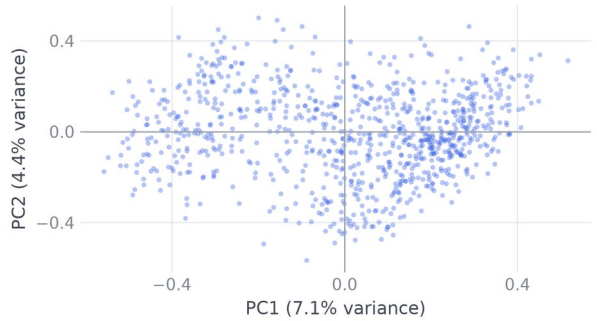
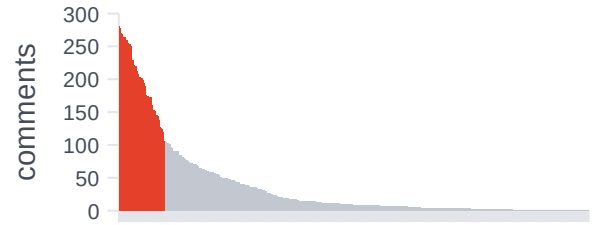
🤔 Why not random?

October comments would leak into training.
The tool only knows the past, **so we test it that way.**

Key point — **Real, recent comments; split by time so the model never sees the future (L06).**

EDA: 3 findings, each changed one step

What we saw → the evidence → what we changed

	1 Wording → 04 Topics	2 PCA → 04 Topics	3 Viral → 05 Score
Finding	**Same meaning, different words**	**No clear groups**	**One video fakes a trend**
Evidence	*"Is this really free?"* *"Do I have to pay for it?"* **TF-IDF · 0 shared words**: **0** unrelated **MiniLM · meaning (L09)**: **0.54** similar 59% of 16,523 training comments: closest in meaning, almost no shared words (MiniLM, L09).	 Example: AI Career, 6,170 comments. No clear groups. Silhouette score 0.02–0.09:	 **53%** of the 16,523 training comments sit under just 10% of the videos A popular video can look like a trend, so counting comments would mislead.
Change	**→ Group by meaning (MiniLM)**	**→ No natural K: choose it in 04**	**→ Count people, max 3 per video**

Key point **Group by meaning · K must be chosen, not found · count people, not comments.**

Find questions (supervised) ①: labels, the answer key

DATA 51,447 raw › 25,226 clean ▸ FILTER 493 labels › 356 train · 137 test › 3,678 questions ▸ TOPICS 2,574 Jul–Sep · 1,104 Oct › 50 topics ▸ SCORE 3 videos

1 What is a question?

Question (1) = the viewer asks for something a video could give

@viewer 1 month ago — 1 · requests
nice man please make a complete roadamap video for Machine Learning Engineer
3 Reply

@viewer 5 days ago — 1 · has a problem
oh, I cannot find GitHub MCP server in Extensions :*(((
Reply · 1 reply

@viewer 2 months ago — 1 · asks
Become AI engineer with out coding background?
1 Reply · 1 reply

@viewer 3 months ago — 0 · just praise
Amazing video on how to transition to AI engineering!
1 Reply

2 How many labels?

493 comments to label 1 or 0

↓ ① picked at random

Random code · make_round1_sample.py

```
# pick at random, fixed seed
train = (pool_train.groupby("niche")
           .sample(120, random_state=42))
test  = pool_test.sample(140, random_state=42)
```

↓ ② split by time (28 Sep)

Train 356	74 questions (21%)	Jul–Sep · ~120 per niche
Test 137	19 questions (14%)	Oct

3 Few labels: 2 checks

Check 1 · Unstable scores
If so: one model looks better just by luck

① CV — gap < SD of 5 fold scores → could be luck — slide 6

Check 2 · Works on new data?
If so: good on training comments, bad on new ones (overfitting)

① test — answered only by the unseen October test — slide 7

Both checks: confirm on the unseen October test — slide 7

Key point 493 labels teach and test the filter.

Find questions (supervised) ②: how we grade

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1 Score: F0.5 + a floor

What do we measure?

Only 21% are questions

21% | 79% others

↓ causes

Class imbalance (L03): accuracy misleads

↓ so use

Precision
kept ones are real?

Recall
found all real ones?

↓ fakes inflate a topic, so precision first

TO RANK **F0.5** precision counts ×2

2 Exam: 5-fold CV

How do we test fairly? (356 training labels)

1	**grade**	learn	learn	learn	learn
2	learn	**grade**	learn	learn	learn
3	learn	learn	**grade**	learn	learn
4	learn	learn	learn	**grade**	learn
5	learn	learn	learn	learn	**grade**

each fold ≈ 71, **15 questions** (21%, stratified)

↓

average = **CV score** → pick the model

so the real test can wait:

October test closed until slide 7

3 Bar: baseline

What must a model beat?

Rule: "?" = a question

	actually yes	actually no
says yes	40	26
says no	34	256

Precision 40/66 = 0.61

Recall 40/74 = 0.54

$$F0.5 = (1 + 0.5^2) \times P \times R / (0.5^2 \times P + R)$$

Baseline F0.5

0.59

every model must beat it

Key point Rank by F0.5 · pick by 5-fold CV · beat the "?" rule (0.59).

Find questions (supervised) ③: features, then model

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1 Features: turn comments into numbers

all scores: CV on the 356 training labels — Baseline F0.5 0.59

0.61 — **Word counts**
TF-IDF (L09), keeps how / what / can

0.45 — **Meaning**
all-MiniLM-L6-v2 (L09)
Why worse? It sees the topic, not the intent.

0.64 — **Meaning + 3 rules ✓**
Output Dimension 384 + 3 joined = 387

the 3 rules are: simple yes/no signs of a question

"?"	how	please
question mark	how / what / can	please / help

2 Pick the model

Model	Precision	Recall	CV F0.5	CV gap vs 0.59
Baseline: "?" only	0.61	0.54	**0.59**	–
Logistic regression	0.75	0.41	**0.64**	+0.05
Random forest	0.79	0.30	**0.59**	±0
Gradient boosting ✓	**0.75**	**0.49**	**0.68**	**+0.09**

↓ tuned by grid search (L04): lr × depth · **default 0.65 → 0.68** (+0.03, small)

5-fold CV: 5 scores of GB → **only 3 of 5 beat the baseline**



✓ **Check 1 · better or just lucky? → impact: small gaps only**
gap 0.09 < SD 0.10 → CV can't tell → **decide on the test (slide 7)**

Key point — **GB was best by CV (0.68 vs 0.59), but within ±0.10: the final test decides.**

Find questions (supervised) ④: the final exam

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3 Final exam on test data

TRAIN
Jul–Sep · 356 labels · learn + CV

TEST
Oct · 137

locked during training → **opened once, here** · 19 real questions inside

Gradient boosting ●●●●●●●●●●●●●○○○○○○○

caught 13 of 19 · 5 false alarms

"?" rule ●●●●●●●●●○○○○○○○○○

caught 10 of 19 · 13 false alarms

● caught ○ missed

Gradient boosting F0.5 **0.71** VS **"?" rule** F0.5 **0.45**

→ **Use gradient boosting**: reads all 25,226 comments, keeps **3,678 questions**

✓ Answers to slide 4: small labels

The 2 checks SD = how much the 5 CV scores swing

Check 1 · Better, or just lucky?

🤔 slide 4

⏳ slide 6: gap 0.09 < SD 0.10

✅ **test: gap 0.26 > SD 0.10 → better**

Check 2 · Works on new data?

🤔 slide 4

✅ **CV 0.68 → test 0.71: yes**

↓ why

Train vs test: what does the model do?

TRAIN · 356 with answers
learns
MiniLM meaning + 3 rules → question?

TEST · 137, answers hidden
learns nothing
applies train's model

Both use MiniLM; only train learns.
Pre-trained (more than 1 billion training pairs) → 356 labels were enough.

Key point **October exam: gradient boosting wins (F0.5 0.71 vs 0.45).**

Topics (unsupervised) ①: group by words or by meaning?

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1 Group 2,574 questions: 4 ways

By words baseline
TF-IDF: same words → same group
short questions share few words → 1 leftover group (AI Tools: 24%)

By meaning L09
MiniLM: same meaning → same group
"Can I do the same with Claude?" ≈ "What tier Claude would I need?"

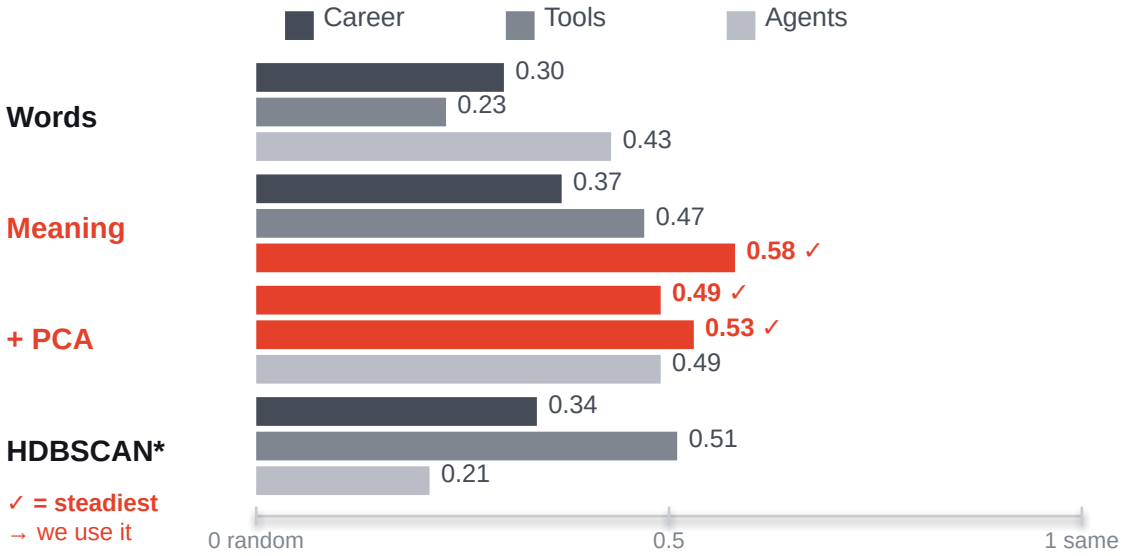
By meaning + PCA L05
384 → ~90 numbers, 80% of info kept
less noise before grouping

HDBSCAN*
finds crowds by itself
but leaves 15–32% of questions in no group

* beyond the course

2 Which way is steadiest? (Adjusted Rand Index, L05)

ARI = do repeated runs give the same groups?
Run each way 5 times with different random starts; compare every pair of runs
0 = no better than random · 1 = identical groups · higher = steadier



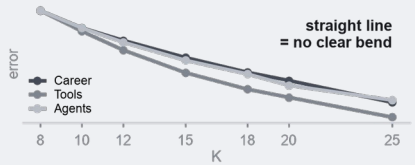
Key point Grouping by meaning was steadiest in all 3 content lines (✓), so that is what we use.

Topics (unsupervised) ②: how many topics? Choose K

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1 Natural K in the data?

Elbow a natural K shows as a bend



✗ **straight line: no bend**

Silhouette a natural K scores near 1

✗ **ours 0.03–0.09**

−1 wrong · 0 overlap · 1 clear

No: the data can't tell us K
→ we must choose K ourselves

2 So I set 2 rules for K

The data can't pick K, so **I set 2 rules**:
each topic must work as 1 video idea

Rule ① 1 video can't cover too much: ≤ 12%

biggest topic ≤ 109 of 912 questions (12%) → else mixed

K = 8 ✗ 16%
143 of 912 questions → too mixed

K = 15 ✓ 11%
97 questions → 1 video

12%

Why 12%? my choice: ≈ 2× an average topic (1/15 = 6.7%)

Rule ② Not too many: 10–25 topics
my choice: a creator can read the list each week

3 Result: K per line

AI Career · 912 questions → cut → **K = 15 topics** → **1 topic = 1 video idea**

e.g. "Becoming an AI engineer without a coding background"
97 questions → 1 video

Break a rule → ✗ out · rest → highest silhouette

K tried	8	10	12	15	18	20	25
Career	✗	✗	✗	**.044**	.028	.044	.040
Tools	✗	✗	✗	.063	✗	**.069**	.063
Agents	✗	✗	.055	**.059**	.052	.039	.054

✗ broke a rule · number = silhouette · **red = chosen K**

Career: 15 ≈ 20 (0.0438 vs 0.0437), almost a tie

15 + 20 + 15 = 50 topics
Career · Tools · Agents → 50 video ideas to score

Key point No natural K in the data, so we chose K so each topic = 1 video: 50 topics.

Topics (unsupervised) ③ follow-up check: fit October?

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1 Draw a ring around each topic (Jul–Sep)

TRAIN **draw on 2,574 Jul–Sep questions**

Example



T-shirt sizes: fit 90% of last year's customers → **this year's still fit?**

→ **Our case: rings fit 90% of Jul–Sep → does October still fit?**



● = 1 Jul–Sep question

least typical 10%
left out

① **Centre** = average of the topic
Centroid · L05

② **Distance** = how similar to the centre
Cosine · L09

③ **Ring** = holds the 90% nearest
10th percentile

Jul–Sep = 90% inside, by design → our reference line
Why 90%? 100% = too loose (everything fits) · 50% = too tight

2 Do October questions land inside?

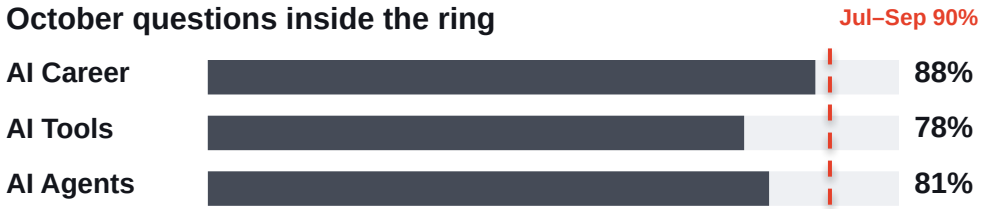
TEST **check 1,104 Oct questions** = 3,678 − 2,574

Each October question → nearest topic centre

● inside = old topic ✓ ○ outside = new theme

K-Means predict

October questions inside the ring



78–88% ≈ 90%: most October questions are inside
↓ **viewers keep asking about the same things**
↓ **the 50 topics still work for new weeks: no rebuild**
check weekly: below 75% → too many new themes → re-cluster

Key point **Viewers keep asking about the same things (78–88% inside vs 90%) → keep the 50 topics.**

Score: who asks, not yet answered, rising

Every Monday: each topic gets 3 signals → one score → Top 3 per content line

1 Score each topic

Example: **"Data science & ML career roadmaps"** · AI Career

Demand	Gap	Fresh	Score
people ask	no video yet	rising	
83	**67**	**58**	**72**
× 0.45 weight	× 0.35 weight	× 0.20 weight	

rank 0–100 = how many of the other 12 Career topics it beats
weight = how much each counts: 45% + 35% + 20% = 100% (our choice)

→ **#1 of 13 AI Career topics: film it**

2 Gap: is there already a video? (L09 Semantic search)

viewer comment ↓ MiniLM → meaning numbers

video title (90 days) ↓ MiniLM → meaning numbers

→ **cosine: how close?**

≥ 0.60	< 0.60
a video already answers it	no video yet = gap

0.60 = the cut that worked best (F1) on 150 checked comment–title pairs

Gap: each question vs. video titles — no match by meaning (< 0.60) = gap

Key point Asked by many, little video yet, rising → this week's #1 in AI Career scores 72.

Back-test (L06): are our top picks still hot a month later?

Pretend it is a past Monday, pick with our score, then check the next 4 weeks

1 How we test (L06: past → future, no peeking)

① **7 past Mondays** (3 Aug – 14 Sep): use only data before that day — Time split · L06

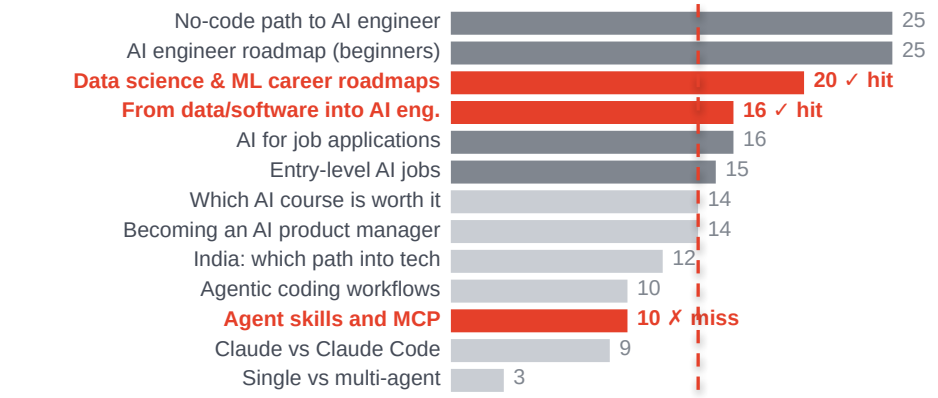
② **Our score picks Top 3 in each line** → 7 × 3 lines × 3 = 63 picks

③ **Wait 4 weeks:** count different people asking each topic (1 person = 1)

④ **Hot?** more people than half the topics in its line → hit ✓ — our rule

2 What is "hot"? A real example

3 Aug · AI Career: people asking in the next 4 weeks



red = our 3 picks → 2 hits of 3 ┆ = half the topics (14): right of it = hot

3 Result: add up all 7 Mondays → our 63 picks vs random

Random — Baseline · L06
3 topics by chance
≈ 30 hits / 63
47%

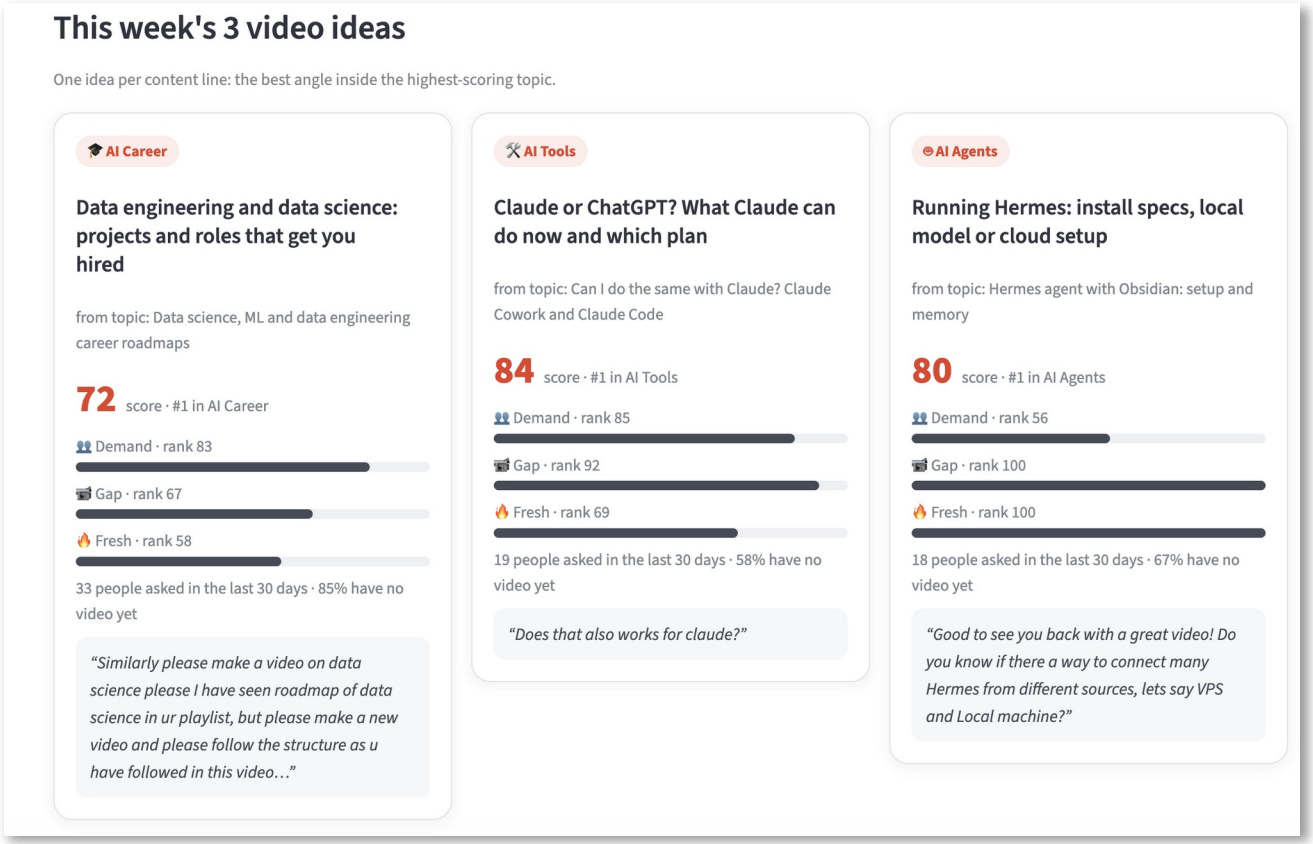
Ours
picked by our score
44 hits / 63
70%

Ours beats random
7 in 10 picks still hot a month later (random: under 5 in 10)
final check 28 Sep: ours 67%, random 49%

Key point — **Ours: 44 of 63 picks (70%) were still hot a month later, vs 47% at random.**

Go live: the dashboard, its value, and what comes next

Streamlit dashboard: next week's 3 videos + 50 K-Means topics



This week's 3 video ideas

One idea per content line: the best angle inside the highest-scoring topic.

AI Career

Data engineering and data science: projects and roles that get you hired

from topic: Data science, ML and data engineering career roadmaps

72 score · #1 in AI Career

Demand · rank 83

Gap · rank 67

Fresh · rank 58

33 people asked in the last 30 days · 85% have no video yet

"Similarly please make a video on data science please I have seen roadmap of data science in ur playlist, but please make a new video and please follow the structure as u have followed in this video…"

AI Tools

Claude or ChatGPT? What Claude can do now and which plan

from topic: Can I do the same with Claude? Claude Cowork and Claude Code

84 score · #1 in AI Tools

Demand · rank 85

Gap · rank 92

Fresh · rank 69

19 people asked in the last 30 days · 58% have no video yet

"Does that also works for claude?"

AI Agents

Running Hermes: install specs, local model or cloud setup

from topic: Hermes agent with Obsidian: setup and memory

80 score · #1 in AI Agents

Demand · rank 56

Gap · rank 100

Fresh · rank 100

18 people asked in the last 30 days · 67% have no video yet

"Good to see you back with a great video! Do you know if there a way to connect many Hermes from different sources, lets say VPS and Local machine?"

Limit · opportunity · next step

LIMIT **Few labels**

Label more

OPPORTUNITY **vs vidIQ / comment tools**

they see searches or one video; mine finds what viewers across channels ask that no video answers, back-tested

SIZE (my time) **Saves 2.5 h a week = 130 h a year**

build ~50 h + 2 h a month upkeep → pays back in ~6 months

NEXT **8-week A/B test**

A: most asked · B: ours · metric: 7-day views · roll out if B ≥ 1.1× A